EXPLICIT HAMILTONIAN FULL PACKINGS OF THE FOUR-BALL BY EQUAL BALLS

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ABSTRACT. For every $k \geq 10$ and $\varepsilon \in (0,1)$ we construct an explicit smooth compactly supported Hamiltonian $H_{k,\varepsilon}$ on $\mathbb{R} \times \mathbb{R}^4$ whose time-one map carries the k closed unit balls centred at $(3i-3,0,0,0)$ into the ball of radius $R = (k/(1-\varepsilon/4))^{1/4}$, filling the fraction $1-\varepsilon/4$. Such packings were previously known to exist only through Biran's theorem, whose proof does not produce the embeddings, and explicitly only for square k. This answers the mathematical part of Epoch AI's construction problem. The two main ingredients are a linear comparison function for which the increasing rearrangement of the profile defect is exactly linear, in place of a contraction fixed point, and a return from the Fermat cubic disc-bundle model to the ball, in which a cube-root symmetry proves avoidance of the line at infinity. The construction brings to dimension four the twisted disc-bundle mechanism that a 2026 preprint of OpenAI introduced in dimensions $2n \geq 6$, and uses neither J-holomorphic curves nor Seiberg–Witten theory. We also prove robustness under perturbation with an explicit margin. The main theorems are verified in Lean 4. General capacities, $k = 9$, higher dimensions and an algorithm with finite output are treated in a companion paper.

1. INTRODUCTION

1.1. **The result.** Points of $\mathbb{R}^{2n} = \mathbb{C}^n$ are $z = (z_1, \dots, z_n)$, $z_j = x_j + iy_j$, with symplectic form $\omega_0 = \sum_j dx_j \wedge dy_j$. The Hamiltonian vector field of a function $H(t,z)$ is defined by $\iota_{X_H}\omega_0 = dH$, i.e. $X_H = (\partial_y H, -\partial_x H)$, and ϕ_H^t denotes its flow from time 0 to time t. We write $B(c) = \{\pi|z|^2 \leq c\}$ for the closed ball of capacity c, $B^\circ(c)$ for its interior, and $B(s,c) = s + B(c)$. Thus $B(s,\pi)$ is the closed unit ball about s, and $B^\circ(\pi R^2)$ is the open Euclidean ball of radius R about the origin.

Theorem A (Equal balls in the four-ball)**.** *Let $k \geq 10$ be an integer and $\varepsilon \in (0,1)$. Put*

$$R = \left(\frac{k}{1-\varepsilon/4}\right)^{1/4}, \qquad C = \pi R^2, \qquad s_i = (3i-3,0,0,0) \in \mathbb{R}^4 \quad (1 \leq i \leq k),$$

and $H_{k,\varepsilon} = H[2,k,C,(\pi,\dots,\pi),(s_1,\dots,s_k)]$ (Definition 7.16). Then $H_{k,\varepsilon}$ is C^∞ on $\mathbb{R} \times \mathbb{R}^4$, vanishes for $t \notin (0,1)$ and outside $\mathbb{R} \times \mathcal{S}$ for an explicit compact set $\mathcal{S}$, and satisfies $\mathrm{Pack}(2,k,C,(\pi,\dots,\pi),s)$ *(Definition 2.1): its flow is global, and its time-one map sends each closed unit ball $B(s_i,\pi)$ into $B^\circ(\pi R^2)$, with pairwise disjoint images. Moreover $k/R^4 = 1-\varepsilon/4 > 1-\varepsilon$.*

Proof. The centres are at distance at least 3, so the closed unit balls are disjoint. Since $R^4 = k/(1-\varepsilon/4) > k \geq 10$, we have $k\pi^2 < \pi^2 R^4 = C^2$ and $3\pi < \pi R^2 = C$, so $(2,k,C,(\pi,\dots,\pi))$ is admissible (Definition 2.2). The normalised capacity $\rho = \pi/C = R^{-2}$ satisfies $\rho\sqrt{k} = \sqrt{1-\varepsilon/4} > \sqrt{3}/2 > \frac{3}{4}$, so the data lie in the linear branch, which uses the linear comparison datum of Proposition 4.4. The proof is completed in §7.4. □

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The Hamiltonian is defined by the same formulas for all (k, ε). It is built from closed formulas, integrals, time-one maps of explicit vector fields, inverses of explicit injective maps and extrema over explicit compact sets, the operations of level L1 in Definition 2.5 (Table 2).

The companion paper [Son26a] defines $H[n, k, C, c, s]$ for all admissible data and proves the packing for unequal capacities in dimension four ($\sum_i c_i^2 < C^2$, $3c_i < C$), for $k = 9$ equal balls and in every dimension $2n \geq 6$; it also gives a uniform algorithm whose output is a rational polynomial times a fixed cutoff. The two papers supersede the single paper [Son26b].

The Epoch problem. Epoch AI [Epo26] asked for an algorithm that, on input $k \geq 10$ and $\varepsilon \in (0, 1)$, outputs an explicit smooth Hamiltonian on $\mathbb{R}^4$ whose time-one map sends the closed unit balls centred at $(3i-3, 0, 0, 0)$ into a closed ball $B^4(R)$ about the origin with $k/R^4 > 1-\varepsilon$. Theorem A and Proposition B answer the mathematical part of this construction problem; the algorithm with finite output is in [Son26a]. Epoch also asks for a PyTorch implementation that a numerical verifier integrates within fixed time limits; the construction here is designed for proof, not for fast integration. A numerical program for that interface, built from the elementary constructions of the third paper of the release ("Elementary packings of the four-ball and their limits"), covers a range of (k, ε) with numerically checked correctness; see the repository.

1.2. **What was known.** Balls of capacities $c_1, \dots, c_k$ embed disjointly in $B^\circ(C) \subset \mathbb{R}^4$ only if $\sum_i c_i^2 < C^2$ and, by Gromov's two-ball theorem [Gro85], $c_i + c_j < C$. McDuff and Polterovich [MP94] related such packings to blow-ups of $\mathbb{CP}^2$ and found the maximal densities for at most nine equal balls; Biran [Bir97] proved that k balls of capacity c embed in $B^\circ(C)$ whenever $k \geq 9$ and $kc^2 < C^2$; and the work of McDuff, Li–Liu and Li–Li [McD91, McD98, LL01, LL02] reduces every four-dimensional ball packing problem to inequalities indexed by exceptional classes. Packing stability was extended by Biran [Bir99], Buse, Hind and Opshtein [BH11, BH13, BHO16] and Edtmair [Edt25], and fails for some open domains [CGH25]. These proofs use J-holomorphic curves and, through Taubes' theorem and inflation, Seiberg–Witten theory; they show that embeddings exist but do not produce a Hamiltonian moving given balls. Explicit packings were known in special cases, and explicit full packings by $k \geq 10$ equal balls only for square k: Traynor's full packings by d^2 equal balls [Tra95], folding, wrapping and the cases $k \leq 9$ [Sch05b, Sch05a, Sch18], tunnelling [Wie09], singular curves and polarisations [Ops15, Ops13], and almost toric methods [Sym03, CV22]. The OpenAI preprint [Ope26] proved the packing criterion conjectured by Siegel and Yao [SY25] in dimensions $2n \geq 6$; in dimension four the two inequalities are not a criterion. We use its twisted disc-bundle mechanism (whose comparison and surface lemmas allow fibres $\mathbb{C}^m$, $m \geq 1$) and the deformation-to-the-normal-cone scheme of its return. Dimension four needs a different model and new arguments: the Fermat cubic disc bundle, the linear comparison identity, which replaces the fixed point by a closed formula, and the symmetry argument that brings the model back to the ball.

The proof here uses explicit maps, forms and flows only, and none of the existence results above. It gives explicit full packings of B^4, i.e. packings of every density below 1, by $k \geq 10$ equal balls without J-holomorphic curves or Seiberg–Witten theory. The companion paper [Son26a] adds the scalar comparison datum, which covers $k = 9$ and unequal capacities, and so proves in the same way Biran's theorem [Bir97] that B^4 admits full packings by k equal balls for every $k \geq 9$; the earlier proofs [Bir97, Bir99, BH13, BHO16] use these tools, directly or through the results they cite.

1.3. **The key ideas.** The model is Biran's decomposition of $\mathbb{CP}^2$ along a smooth cubic E [Bir01]: the complement of an isotropic skeleton is the degree-9 disc bundle over E with fibres of capacity $\frac{1}{3}$, and its base area at fibre moment p is $A(p) = 3(1-3p)$. Balls are placed fibrewise; for $k \geq 10$ balls of capacity close to $1/\sqrt{k}$ the volume budget holds but the slice profile $\sum_i (a_i - p)_+$ exceeds

$A(p)$ near $p = 0$ (Figure 1). Following [Ope26], the bundle is twisted across an annulus in the base by an area-preserving map q of the fibre disc. Paired with a function K of the moment, the twist moves base area between moment levels, and the balls fit if, for $P = A - \sum_i (a_i - p)_+$ and some gap $\gamma > 0$,

$$(K - P)\big(p(q(w))\big) \le K\big(p(w)\big) - \gamma \qquad \text{for every fibre point } w$$

(Section 3). In [Ope26] the function K is the fixed point of a contraction built from increasing rearrangements, and the planar maps realising the rearrangements and the relative Moser isotopies are obtained from existence statements. Two ideas make the four-dimensional construction explicit.

The linear comparison identity (§4.2). For $k \ge 10$ equal balls of capacity $a = t_0/\sqrt{k}$, with the bundle cut at fibre moment $h = t_0/3$, the function $K - P$ with K linear of slope $3\sqrt{k} - 9$ is, up to the constant $-g$, a multiple of a tent Λ with $\Lambda(0) = \Lambda(h) = h$. Because the tent reaches the same height at both ends, each sublevel interval $\{\Lambda \le b\}$ has length exactly b, so the increasing rearrangement of Λ is the identity and the comparison inequality reduces to the constant gap $g = 3(1 - t_0)$. One explicit planar map realises the rearrangement at a cost of at most $g/32$. No fixed point is needed.

The return to the ball (§6.2). The disc bundle is written by theta functions on the cover $\mathbb{R}^2 \times \mathbb{C}$ of the Fermat torus and carried into $\mathbb{CP}^2$ by explicit injection, transport and Moser flows. The ball is $\mathbb{CP}^2$ minus the line $\{Z_0 = 0\}$, so the images must avoid this line. All the return maps commute with the cube-root symmetry $Z_0 \mapsto e^{2\pi i/3} Z_0$. A point mapped into the line is fixed by the symmetry, hence comes from a point with $Z_0 = 0$, and the handle construction keeps the balls away from such points. Avoidance thus follows from symmetry, not from an estimate.

1.4. **Organisation.** Section 2 states the packing specification, the robustness estimate and the form of Theorem A for the Epoch data. After the dilation by $1/\sqrt{C}$ the construction has five steps. (1) *Model:* the cubic disc bundle on the cover of the Fermat torus, cut at fibre moment $p \le h$ (§3, §6.1). (2) *Comparison:* the linear K and an explicit fibre isotopy q_t with gap $\frac{31}{32} g$ (Section 4). (3) *Handle:* the twist and three explicit Moser flows place the balls in disjoint layers over an annulus (Section 5). (4) *Return:* explicit flows carry the model into $B^\circ(1)$ (Section 6); Table 1 lists the margins passed from step to step. (5) *Motion:* one isotopy from the given balls to the embeddings, disjoint at all times, and the Hamiltonian generating it (Section 7). Section 8 describes the Lean 4 formalisation. The supplement contains the complete proofs of the statements proved here only in outline; numbers with the prefix S, as in §S5, refer to it.

Conventions. The dilation $z \mapsto \lambda z$ multiplies ω_0 and capacities by λ^2. In fibre coordinates w_j the moments are $p_j = \pi |w_j|^2$ and the angles $\theta_j = \arg w_j / 2\pi \in \mathbb{R}/\mathbb{Z}$, so the fibre form is $\omega_V = \sum_j dp_j \wedge d\theta_j$. Fibre Hamiltonians Q follow the sign of [Ope26], $\iota_{X_Q} \omega_V = -dQ$, opposite to the global one. In affine coordinates $\omega_{\mathrm{FS}} = dd^c \log(1 + |z|^2)$ with $d^c = i(\bar\partial - \partial)/(4\pi)$; the map $z \mapsto z/\sqrt{\pi(1 + |z|^2)}$ identifies this chart symplectically with $B^\circ(1)$.

2. Statements

2.1. The packing specification.

Definition 2.1 (Packing specification)**.** Let $n \ge 1$, $k \ge 0$, $C > 0$, $c = (c_1, \dots, c_k) \in (0, \infty)^k$ and $s = (s_1, \dots, s_k) \in (\mathbb{R}^{2n})^k$. A function $H \colon \mathbb{R} \times \mathbb{R}^{2n} \to \mathbb{R}$ satisfies $\mathrm{Pack}(n, k, C, c, s)$ if

(a) H is C^∞, and there is a compact set $\mathcal{S} \subset \mathbb{R}^{2n}$ with $H(t, z) = 0$ for all $t \in \mathbb{R}$ and all $z \notin \mathcal{S}$;

(b) for every i and every $z \in B(s_i, c_i)$ there is an integral curve $\gamma\colon \mathbb{R} \to \mathbb{R}^{2n}$ of X_H with $\gamma(0) = z$, and every integral curve $\gamma\colon [0,1] \to \mathbb{R}^{2n}$ of X_H with $\gamma(0) = z$ satisfies $\gamma(1) \in B^\circ(C)$.

For H satisfying (a) the flow ϕ_H^t is global and consists of symplectic diffeomorphisms, and (b) says $\phi_H^1(B(s_i, c_i)) \subset B^\circ(C)$ for every i.

Definition 2.2 (Admissible data)**.** Let $n \geq 2$, $k \geq 1$, $C > 0$ and $c \in (0,\infty)^k$. The data (n, k, C, c) are *admissible* if

$$\begin{aligned} n \geq 3: &\qquad \textstyle\sum_i c_i^n < C^n \quad \text{and} \quad c_i + c_j < C \ \text{ for all } i \neq j; \\ n = 2: &\qquad \textstyle\sum_i c_i^2 < C^2 \quad \text{and} \quad 3c_i < C \ \text{ for all } i. \end{aligned}$$

For $n = 2$ the condition $3c_i < C$ comes from the fibre capacity $\frac{1}{3}$ of the model; it is a sufficient small-capacity range, not the full four-dimensional packing criterion. This paper uses admissibility only for $n = 2$ and equal capacities.

2.2. **Robustness.** Theorem A leaves an explicit margin: the images lie in the closed ball of radius R, while the density requirement is already met by a larger radius R'. The following estimate converts this margin into a tolerance on the vector field.

Proposition B (Robustness)**.** *Let $k \geq 10$, $\varepsilon \in (0,1)$, $R = (k/(1-\varepsilon/4))^{1/4}$ and $R' = (k/(1-\varepsilon/2))^{1/4}$, so $R < R'$. Let $\widehat{H}$ be a smooth function on $\mathbb{R} \times \mathbb{R}^4$ vanishing outside $\mathbb{R} \times K$ for a compact set K. Suppose that, for $t \in [0,1]$, the field $X_{\widehat{H}}(t,\cdot)$ is L-Lipschitz for some $L \geq 0$ and $|X_{\widehat{H}} - X_{H_{k,\varepsilon}}| \leq \eta$ on $\mathbb{R}^4$, where $\eta e^L \leq R' - R$. Then the time-one map of $\widehat{H}$ sends each closed unit ball $B(s_i, \pi)$ into the closed ball of radius R' about the origin, and $k/R'^4 = 1 - \varepsilon/2 > 1 - \varepsilon$.*

Proof. Let γ and $\hat{\gamma}$ be integral curves of $X_{H_{k,\varepsilon}}$ and $X_{\widehat{H}}$ starting at the same point of $B(s_i, \pi)$. Then $|\hat{\gamma}(1) - \gamma(1)| \leq \eta(e^L - 1)/L \leq \eta e^L$ by Grönwall's inequality (Lemma 2.3, with the Lipschitz bound on $X_{\widehat{H}}$), and $|\gamma(1)| \leq R$ by Theorem A. For $L = 0$ the Grönwall bound is η. □

No explicit bound on the Lipschitz constant of $X_{H_{k,\varepsilon}}$ is given, so the margin $R' - R$ is not directly a numerical step-size bound.

Lemma 2.3 (Grönwall)**.** *Let $G, H\colon \mathbb{R} \times \mathbb{R}^{2n} \to \mathbb{R}$ be C^∞, let $\eta > 0$ and $L_X > 0$, and suppose that*

(i) $|X_G(t,z) - X_H(t,z)| \leq \eta$ for all $t \in [0,1]$ and all $z \in \mathbb{R}^{2n}$;
(ii) $X_H(t,\cdot)$ is L_X-Lipschitz on all of $\mathbb{R}^{2n}$ for every $t \in [0,1]$.

If $\gamma_G, \gamma_H\colon [0,1] \to \mathbb{R}^{2n}$ are integral curves of X_G and X_H with $\gamma_G(0) = \gamma_H(0)$, then $|\gamma_G(1) - \gamma_H(1)| \leq \eta(e^{L_X} - 1)/L_X$.

Proof. $u = \gamma_G - \gamma_H$ satisfies $u(0) = 0$ and $|\dot{u}| \leq \eta + L_X|u|$, so Grönwall's inequality gives the bound. □

For smooth compactly supported H, (ii) holds for every $L_X > 0$ with $L_X \geq \max_{[0,1]\times\mathbb{R}^{2n}} \|D_z^2 H\|$, as $X_H = J_0 \nabla_z H$ with J_0 orthogonal.

2.3. **The Epoch data.**

Corollary C (The Epoch data)**.** *Let $k \geq 10$ be an integer, $\varepsilon \in (0, \frac{7}{8})$, $s_i = (3i-3, 0, 0, 0) \in \mathbb{R}^4$, $R = (k/(1-\varepsilon/2))^{1/4}$ and $C = \pi R^2$. Then $H = H[2, k, C, (\pi, \dots, \pi), s]$ satisfies* $\mathrm{Pack}(2, k, C, (\pi, \dots, \pi), s)$*: its time-one map sends each closed unit ball $B(s_i, \pi)$ into $B^\circ(\pi R^2)$, and $k/R^4 = 1 - \varepsilon/2 > 1 - \varepsilon$.*

Theorem A is Corollary C at $\varepsilon/2$. The bound $\varepsilon < \frac{7}{8}$ is the range $\rho\sqrt{k} = \sqrt{1-\varepsilon/2} > \frac{3}{4}$ of the linear comparison datum. The companion paper [Son26a] proves the corollary for all $k \geq 9$ and $\varepsilon \in (0,1)$, with the scalar comparison datum outside this range, and uses it as the input of its finite-output algorithm.

2.4. **Explicitness.**

Definition 2.4 (Definite description)**.** For a property $\mathcal{P}$ of elements of a set X and a proof h that exactly one $x \in X$ satisfies $\mathcal{P}(x)$, the(h) denotes that element. Phrases such as "the unique u with $F(u) = y$" denote this operation.

Definition 2.5 (Explicit definitional construction)**.** An *explicit definitional construction* (level L1) is a definition built from rational constants and the input by arithmetic and elementary functions, integrals over compact domains, limits of series with an explicit rate of convergence, maxima and minima of continuous functions over explicit compact sets, unique solutions of explicit ordinary differential equations, inverses of explicit injective maps, unique roots of explicit equations, case distinctions on propositions about the input and about the arguments of the function being defined, and least elements of searches over a fixed enumeration of finitely described objects (integers, rationals, and tuples of these and of rational polynomials) whose validity is decided by finitely many explicit inequalities; definite descriptions are used only after unique existence has been proved (Definition 2.4).

Theorem A concerns level L1 (Table 2). The finite form and computability (levels L2 and L3) are defined in [Son26a].

3. The mechanism

We follow the handle-packing scheme of [Ope26]. Give $\mathbb{CP}^2$ the Fubini–Study form for which a line has area 1, so that $\operatorname{vol}\mathbb{CP}^2 = \frac{1}{2}$ and the complement of a line is $B^\circ(1)$. Let $E \subset \mathbb{CP}^2$ be a smooth cubic, of area 3. Biran [Bir01] showed that the complement of an isotropic skeleton in $\mathbb{CP}^2$ is symplectomorphic to a disc bundle over E: the normal bundle, of degree $E \cdot E = 9$, with fibre discs of capacity $\frac{1}{3}$. With fibre moment $p = \pi|w|^2$, the base area of the circle bundle at level p is

$$A(p) = 3 - 9p = 3(1-3p) \qquad (0 \leq p < \tfrac{1}{3}), \qquad \int_0^{1/3} A(p)\,dp = \frac{1}{2} = \operatorname{vol}\mathbb{CP}^2.$$

The cubic is the first smooth plane curve with the topology needed by the handle construction: its base must carry a closed one-form with nonzero period around the handle, which $H^1(S^2;\mathbb{R}) = 0$ rules out for a line or a conic. The fibre bound $\frac{1}{3}$, and hence the condition $3c_i < C$, reflects this choice. The target capacity is 1; the general case follows by the dilation of Definition 7.1. The toric balls and the cap integral below are stated for fibres $\mathbb{C}^m$, $m = n-1$; in this paper $n = 2$ and $m = 1$.

3.1. **Toric balls and area profiles.**

On $\mathbb{C}^m$ let $p_j = \pi|w_j|^2$, $|p| = \sum_j p_j$ and $\omega_V = \sum_j dx_j \wedge dy_j = \sum_j dp_j \wedge d\theta_j$ (angles of period one). In dimension four the moment interval is $\Delta = [0,h]$ with $h < \frac{1}{3}$, and $V_\Delta = \{w \in \mathbb{C} : p(w) \leq h\} = \bar{D}(h)$; the source caps stay in $\{p \leq a_i\}$ with $a_i < h$, away from the outer boundary $p = h$.

Definition 3.1 (Toric ball)**.** In $\mathbb{C}^n = \mathbb{C} \times \mathbb{C}^m$, put $\varpi = \pi|z_{\mathrm{b}}|^2$. The ball of capacity a is $B(a) = \{\varpi + |p| \leq a\}$. At a fibre point with $|p| \leq a$, its base slice is a disc of capacity $a - |p|$; its slice height is 0 for $|p| > a$. Thus the slice-height function is $(a-|p|)_+$.

Lemma 3.2 (The cap integral)**.** *For $a \geq 0$,*

$$\int_{\mathbb{R}^m_{\geq 0}} (a - |p|)_+ \, dp = \frac{a^{m+1}}{(m+1)!}.$$

Proof. Integrate $(a-t)_+$ against the distribution function $t^m/m!$ of $|p|$. □

The *base profile* $H_{\mathrm{b}}(p)$ of a model is the base area of the circle bundle at moment level p, and its *budget* over Δ is

$$\mathcal{B}_\Delta = \int_\Delta H_{\mathrm{b}}(p) \, dp. \tag{1}$$

The cubic model has $H_{\mathrm{b}}(p) = A(p) = 3(1-3p)$ on $\Delta = [0,h]$ with $h < 1/3$, so $\mathcal{B}_{[0,h]} = 3h - \frac{9}{2}h^2 \to \frac{1}{2}$; its coordinates and deck maps are given in Section S6.

3.2. **Why stacking fails.** A ball $B(a) \subset \mathbb{C}\times\mathbb{C}$, sliced by the moment p of its second coordinate, is a family of discs of capacity $a - p$ for $p \leq a$, and $\int_0^\infty (a-p)_+ \, dp = a^2/2 = \operatorname{vol} B(a)$. Placing balls fibrewise in the bundle requires $a < \frac{1}{3}$. Choose auxiliary capacities $a_i < h$ larger than the given ones, and set $P_{\mathrm{b}}(p) = H_{\mathrm{b}}(p) - \sum_i (a_i - p)_+$. By the cap integral,

$$\int_0^h P_{\mathrm{b}} = \mathcal{B}_{[0,h]} - \sum_i \frac{a_i^2}{2}. \tag{2}$$

If $P_{\mathrm{b}} \geq 0$, the base discs can be placed side by side over E; for nine balls of capacity $\frac{1}{3}$ this holds with equality. A positive integral does not give $P_{\mathrm{b}} > 0$ pointwise. In the cubic model with $k \geq 10$, $t_0 \in (0,1)$, $a_i = t_0/\sqrt{k}$ and $h = t_0/3$, one has $\int_0^h P_{\mathrm{b}} = t_0(1-t_0) > 0$, while $P_{\mathrm{b}}(0) = 3 - \sqrt{k}\, t_0 < 0$ for $t_0 > 3/\sqrt{k}$ (Figure 1). The handle construction needs only the positive integral.

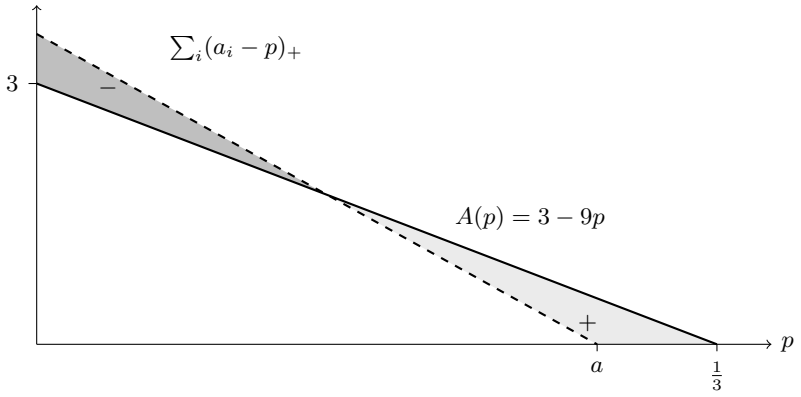


FIGURE 1. The area profile A of the cubic model (solid) and the slice profile of 13 balls of capacity $a = 0.99/\sqrt{13}$ (dashed). The deficit (dark) is smaller than the surplus (light), since $\int_0^h (A - \sum_i (a_i - p)_+) \, dp = \frac{1}{2}(1 - 13a^2 - (1-3h)^2) > 0$ for $h = 0.33$, but stacking needs $A \geq \sum_i (a_i - p)_+$ pointwise. The twist moves area from the surplus to the deficit.

3.3. **The twist moves area.** The construction of [Ope26] pays the deficit with the surplus by twisting the bundle: across an annulus in the base, the fibre over one boundary circle is identified with the fibre over the other by an area-preserving map q of the fibre disc instead of the identity. On the annulus $(s, v) \in [0,1] \times \mathbb{R}/\mathbb{Z}$, let $\beta = b(v)\,dv$ with $b \geq 0$ supported in a band and $\int_0^1 b = 1$, and let ν be the base area form. The first Moser path (Section 5) replaces the model form on the annulus by $\omega_V + e_h\nu + d(f(s)H^*(p)\beta)$, with background $e_h = g/16$, $H^* = H_{\mathrm{b}} - e_h$ and a clock f rising from 0 to 1; the comparison gap is $\frac{31}{32}g$ and the residual is at least $\frac{7}{8}g$ (Lemma S8.3; Table 1).

Let q_t be the comparator's Hamiltonian isotopy of the fibres, $q = q_1$, and $\mathcal{D}_\lambda(s, v, w) = (s, v, q_{\lambda f(s)}w)$. For a smooth function K of the moments put $J = K - H^*$; adding $d(J(p)\beta)$ changes only toric mixed terms off the annulus. After the twist, the area available across the annulus at w is

$$K\big(p(q^{-1}w)\big) - J\big(p(w)\big) = H^*\big(p(w)\big) + K\big(p(q^{-1}w)\big) - K\big(p(w)\big). \tag{3}$$

Since q preserves the fibre area, this redistributes area between moment levels without changing the total. The comparison supplies, for a minorant $P \leq P_{\mathrm{b}}$,

$$(K - P)\big(p(qw)\big) \leq K\big(p(w)\big) - \gamma \qquad (w \in V_\Delta). \tag{4}$$

Integrating over V_Δ shows that this forces $\int_\Delta P > 0$.

The identity (3) is checked directly. The fibre is $\bar{D}(h)$, and over the annulus every form is $\omega_V + e_h\nu + d(G\beta)$ with $G(s, w)$ independent of v. By Lemma 5.1(b), its restriction to the annulus through w is $e_h\nu + \partial_s G\,ds \wedge \beta$. Hence, beyond the background, the area above w is $G(1, w) - G(0, w)$, which equals $H^*(p(w))$ after the first path. The twist $\mathcal{D}$ is the identity for $s \leq 0$ and $(s, v, w) \mapsto (s, v, qw)$ for $s \geq 1$. After adding $d(J(p)\beta)$, the coefficient below the annulus is therefore $J(p(w))$. Above it, the coefficient is $H^*(p(z)) + J(p(z)) = K(p(z))$ at $z = q^{-1}w$, which gives (3).

The *layers* are disjoint subintervals $I_1, \ldots, I_k, I_*$ of $(0, 1)$ (Figure 2). Across I_i the coefficient rises by a smooth cap $h_i(p(w))$. This cap is an explicit smoothing of $(a_i - p)_+$, above it by at most $\frac{g}{32}$ in total, and it holds a ball of capacity slightly less than a_i in product form, since the form is toric on I_i. Across I_* the coefficient rises by the remainder $R^*(w)$. This remainder is positive because γ exceeds the smoothing errors, and it is the only layer coefficient depending on more than $p(w)$.

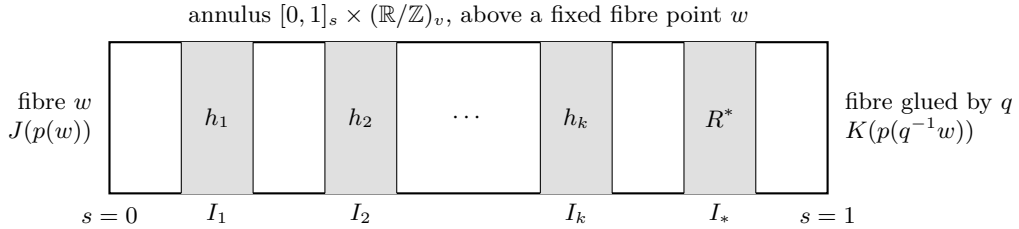


FIGURE 2. The annulus above a fibre point w in dimension four. The β-coefficient rises from $J(p(w))$ to $K(p(q^{-1}w))$; across the layer I_i it rises by the cap $h_i(p(w))$ of the i-th ball and across I_* by the positive remainder $R^*(w)$.

3.4. **The comparison datum.**

Definition 3.3 (Comparison datum)**.** A comparison datum for $([0, h], P)$ with gap $\gamma > 0$ consists of a function K, smooth on a neighbourhood of $[0, h]$, and a smooth Hamiltonian isotopy $(q_t)_{t\in[0,1]}$ of $\mathbb{C}$ with $q_0 = \mathrm{id}$, generated by Q_t with $\iota_{X_{Q_t}}\omega_V = -dQ_t$, such that:

(1) $q_t(\bar{D}(h)) = \bar{D}(h)$ for every t, and there is $\varsigma > 0$ such that $Q_t = 0$ on the collar $\{p > h - \varsigma\}$ for every t;
(2) $q = q_1$ satisfies the comparison inequality (4).

Section 4 constructs the datum in closed form for equal balls with $k \geq 10$ and $\frac{3}{4} < \rho\sqrt{k} < 1$ (Proposition 4.4). Comparison data over polytopes, for general concave profiles, are constructed in [Son26a].

4. Hamiltonian comparison

This section produces a comparison datum (Definition 3.3) for the profile of $k \geq 10$ equal balls. Let $h > 0$, $\bar{D}(h) = \{w \in \mathbb{C} : p(w) \leq h\}$ with $p = \pi|w|^2$, $\theta = \arg w/2\pi$ and $\omega_V = dp \wedge d\theta$ (§3), and let P be continuous and concave on $[0, h]$ with positive mean. We seek a smooth K and a Hamiltonian isotopy q_t of $\bar{D}(h)$ with $(K - P) \circ p \circ q_1 \leq K \circ p - \gamma$ for some $\gamma > 0$. Since $p_*\omega_V$ is Lebesgue measure on $[0, h]$, q_1 only redistributes the values of $K - P$.

The rearrangement operator is that of [Ope26, §2], where K is the fixed point of a contraction built from it. For $k \geq 10$ equal balls, §4.2 gives a closed-form datum with linear K and no fixed point. Everything is explicit at level L1 (Definition 2.5).

4.1. **The increasing rearrangement.**

Definition 4.1 (Increasing rearrangement)**.** For a continuous convex function F on $[0, h]$ put

$$(R_1F)(x) = \min_{0 \leq a \leq h - x} \max\big(F(a),\, F(a + x)\big), \qquad 0 \leq x \leq h.$$

The minimum exists, and since sublevel sets of convex functions are intervals, R_1F is the increasing rearrangement of F on $[0, h]$. The properties used below are in Lemma S3.2: R_1 is monotone, commutes with adding constants and with multiplication by $\alpha \geq 0$, and preserves integrals and maxima.

4.2. **The linear comparison identity.**

For $k \geq 10$ equal capacities in dimension four no fixed point is needed. The height $h = t_0/3$ is chosen so that, for linear K, the function $K - P$ is a constant plus a multiple of a tent that takes the same value at both ends of $[0, h]$. Each sublevel interval of such a tent has length equal to its level, so its increasing rearrangement is the identity, and the comparison inequality reduces to a constant gap. This datum serves Theorem S10.23.

Here $\Delta = [0, h]$ with h real (rationality is not used), so $V_\Delta = \bar{D}(h)$. The symbols s, a, c_s, g, A, B are local to this subsection and to §S3.3. For an integer $k \geq 10$ and $\eta_\star \in (0, \frac{1}{4})$, set $t_0 = 1 - \eta_\star/2$ and

$$s = \sqrt{k}, \quad a = \frac{t_0}{s}, \quad h = \frac{t_0}{3}, \quad c_s = 3s - 9, \quad g = 3(1 - t_0), \quad A = \frac{h}{a}, \quad B = \frac{h}{h - a}.$$

Since $s > 3$, we have $0 < a < h$, $c_s > 0$, $A, B > 0$, and

$$\frac{1}{A} + \frac{1}{B} = \frac{a}{h} + \frac{h - a}{h} = 1. \tag{5}$$

Let $\Lambda(x) = A(a - x)$ for $x \leq a$ and $\Lambda(x) = B(x - a)$ for $x \geq a$. This is a tent with $\Lambda(0) = \Lambda(h) = h$ and $\Lambda(a) = 0$.

Lemma 4.2 (Sublevel sets of the tent)**.** *For $b \geq 0$, $\{x \in \mathbb{R} : \Lambda(x) \leq b\} = [a - b/A,\, a + b/B]$, an interval of length b. If $b \leq h$, it is contained in $[0, h]$.*

Proof. The interval is read off from the two affine pieces. Its length is $b(1/A + 1/B) = b$ by (5). If $b \leq h$, then $a - b/A \geq a - h/A = 0$ and $a + b/B \leq a + h/B = h$. □

Corollary 4.3 (Rearrangement of the tent)**.** *On* $[0,h]$, $R_1\Lambda(x) = x$ *for* $0 \le x \le h$. *Consequently* $R_1(-g + c_s\Lambda)(x) = -g + c_s x$.

Proof. With $\alpha = a - x/A$ we have $\alpha \ge 0$, $\alpha + x = a + x/B \le h$ and $\Lambda(\alpha) = \Lambda(\alpha + x) = x$, so $R_1\Lambda(x) \le x$. Conversely, suppose an admissible $[\alpha, \alpha + x]$ had $b' = \max(\Lambda(\alpha), \Lambda(\alpha + x)) < x$. By convexity it would lie in $\{\Lambda \le b'\}$, which has length $b' < x$ (Lemma 4.2); this is impossible. The second claim follows from Lemma S3.2(iv), since $c_s > 0$. □

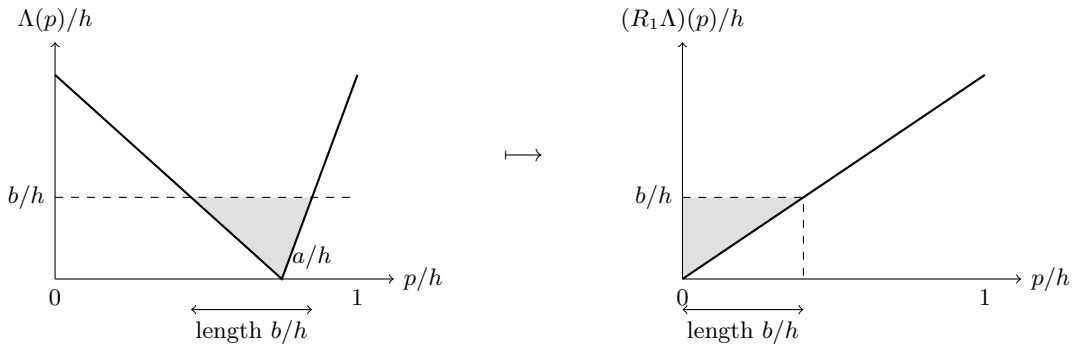


FIGURE 3. The linear comparison: the sublevel interval of Λ has length b, so its increasing rearrangement is p. The drawing uses $a/h = 3/4$ ($k = 16$); the identity holds for every $k \ge 10$. Consequently $K - R_1(K - P) \equiv g$, before the planar realisation error is deducted.

Proposition 4.4 (Linear comparison datum for equal balls)**.** *With the notation above let* $\Delta = [0,h]$, $P(p) = 3(1-3p) - k(a-p)_+$, $K(p) = c_s p$ *and* $F = K - P$. *Then*

$$F(x) = -g + c_s\Lambda(x) \qquad (x \in \mathbb{R}). \tag{6}$$

Let $e_q = \min\big(h/100,\ g/(128c_s)\big)$ *(Definition S3.4) and let* (q_t) *be the isotopy of Theorem S5.34, whose endpoint* $q = q_1$ *satisfies* $\Lambda(p(q(z))) \le \pi|z|^2 + 4e_q$ *on* $\bar{D}(h)$. *Then* $(K, (q_t))$ *is a comparison datum for* (Δ, P) *with gap* $\frac{31}{32}g$.

Proof. By Lemma S3.3, $c_sA = k - 3s$, $c_sB = 3s$ and $3sa = 3 - g$. Hence both sides of (6) equal $(3s-k)x + ka - 3$ for $x \le a$, and $3sx - 3$ for $x \ge a$.

By Theorem S5.34, $q_0 = \mathrm{id}$ and each q_t maps $\bar{D}(h)$ onto itself. Moreover $q_t = \mathrm{id}$ and $Q_t = 0$ on $\{\pi|z|^2 > P_{\mathrm{supp}}\}$ for a constant $P_{\mathrm{supp}} < h$; this gives condition (1) of Definition 3.3.

For condition (2), let $w \in \bar{D}(h)$ and $I = p(w)$. By (6) and $4c_se_q \le g/32$,

$$K(I) - (K-P)\big(p(q(w))\big) = c_sI + g - c_s\Lambda\big(p(q(w))\big) \ge g - 4c_se_q \ge \tfrac{31}{32}g. \qquad \square$$

Thus $K - R_1F \equiv g$ exactly (Corollary 4.3), and the planar map costs at most $g/32$. Ideally q would carry each centred disc $\bar{D}(b)$, $0 \le b \le h$, onto the sublevel set $\{\Lambda \circ p \le b\} = \{a - b/A \le p \le a + b/B\}$, an annulus (a disc for $b = h$) of the same area b, nested in b; no area-preserving homeomorphism does this exactly (§S3.3). The map of Theorem S5.34 is built from Morse charts of a smoothed tent potential. On the core $\bar{D}(h - \frac{31}{8}e_q)$ its error is below $\frac{13}{24}e_q$; the full loss $4e_q$ occurs only on the outer annulus, where q is joined to the identity. In [Ope26] the comparison function is the fixed point of the contraction T_λ and the gap comes from the contraction argument; here the identity (6) gives it directly. As $t_0 \to 1$, $g \to 0$ while $ka^2 = t_0^2 \to 1$. The equal-capacity branch takes $\eta_\star = 1 - \rho\sqrt{k}$, so $t_0 = (1 + \rho\sqrt{k})/2$ and $g = 3\eta_\star/2$.

Remark 4.5 (The range of the linear datum). The linear comparison is used when the capacities are equal, $k \geq 10$ and $\eta_\star = 1 - \rho\sqrt{k}$ lies in $(0, \frac{1}{4})$, that is $\frac{3}{4} < \rho\sqrt{k} < 1$. For $k = 9$ the formulas give $a = h$, and for unequal capacities a linear K can fail; these inputs use the scalar comparison datum of the companion paper [Son26a].

5. Packing over a handle

The handle construction is an explicit version of the surface packing lemma of [Ope26, §3.2], with the Moser argument [Mos65] carried out by explicit vector fields. We describe it for the cubic model; the higher-dimensional version is in [Son26a].

Setting. The base is the torus Γ, and we work on the cover $\mathbb{R}^2 \times \mathbb{C}$ with coordinates (u, v, w), $p = \pi|w|^2$, equivariantly under the deck maps, which shift (u, v) by integers and preserve p (Definition S6.1). The *marked fibres* are the fibres over the three points of Γ where $Z_0 = 0$. The handle data (Proposition S6.3) are an annulus $\mathcal{A} = \{\frac{1}{8} \leq u \leq \frac{5}{24}\} \subset \mathcal{A}^+ = \{\frac{1}{12} \leq u \leq \frac{1}{4}\}$, which misses the marked points, with handle coordinate $s = 12u - \frac{3}{2} \in [0, 1]$; the area form $\nu = du \wedge dv$; and the closed one-form $\beta = b(v)\, dv$ with $b \geq 0$ supported in a band and $\int_0^1 b = 1$. Over $\mathcal{A}^+$ the bundle is trivialised by the frame chart $\chi_{\mathcal{A}}(u, v, w_{\mathcal{A}}) = (u, v, (Z_0/|Z_0|)^3 w_{\mathcal{A}})$ (Definition S8.4). The data $h, g, K, (q_t)$ are valid cubic data (Definition S8.1): in the linear branch, those of Proposition 4.4.

The three paths. All three are exact paths of deck-invariant forms with explicit primitives.

(1) The *first path* $\Omega_t = \Omega_0 + t\, d\alpha_1$, with the closed-form primitive α_1 of Construction S7.23, moves the usable base area onto the annulus: over $\mathcal{A}$, $\chi_{\mathcal{A}}^* \Omega_1 = \omega_V + e_h \nu + d\big(f(u) H^*(p) \beta\big)$ with $e_h = g/16$, $H^* = H_{\mathrm{b}} - e_h$ and the clock f, which rises from 0 to 1 inside $\mathcal{A}$ (Lemma S8.5).
(2) The *second path* inserts the comparator isotopy across the annulus by the twist below (Proposition S8.9).
(3) The *third path* separates the caps h_i of the balls into disjoint layers in the u-direction, with a positive residual layer (Proposition S8.13).

The Moser flow of the concatenated path is the handle Moser map Φ, and the layer maps E_i place the balls in the layers (Remark 5.3).

5.1. The twist.

The two computations behind the second path hold for fibres $\mathbb{C}^m$ of any rank m; we state them in that generality because the companion paper [Son26a] uses them for $m \geq 2$. Here $m = 1$, $n = 2$, $\omega_V = \sum_j dp_j \wedge d\theta_j$ is the fibre form, and the handle coordinates are (s, v). A *horizontal* one-form annihilates fibre tangent vectors.

Lemma 5.1 (Top power). *(a) In local product coordinates let κ' be a base two-form and $a'_1, \dots, a'_m$ base one-forms, with coefficients depending on the base point and on p. Then $(\omega_V + \kappa' + \sum_j dp_j \wedge a'_j)^n = n\, \omega_V^m \wedge \kappa'$.*

(b) On $\mathcal{A}^+ \times \mathbb{C}^m$, in product coordinates, let $\sigma_{\mathcal{A}} = a_{\mathcal{A}}(s, v)\, ds \wedge dv$ be a positive area form on the annulus, let $G(s, w)$ be smooth and independent of v, and $e > 0$. Then $\big(\omega_V + e\sigma_{\mathcal{A}} + d(G\beta)\big)^n = n\, (e a_{\mathcal{A}}(s, v) + b(v)\, \partial_s G)\, \omega_V^m \wedge ds \wedge dv$.

In particular the forms in (a) are nondegenerate where $\kappa' > 0$; when closed, they are symplectic there. The forms in (b) are symplectic where $\partial_s G \geq 0$.

Proof. Let x_j, y_j be the fibre covectors and σ, τ the base covectors at a point. Then $dp_j = u_j x_j + v_j y_j$ and $a'_j = A_j \sigma + B_j \tau$, and $x_j \wedge y_j + dp_j \wedge a'_j = (x_j - v_j a'_j) \wedge (y_j + u_j a'_j)$, so each block is decomposable. The blocks commute and square to zero, so the m-th power of their sum is $m!$ times their product. Every base correction in that product vanishes against the base area. Adding κ' then gives (a), including where a moment differential vanishes. For (b), the mixed

part of $d(G\beta) = \partial_s G\, ds\wedge\beta + d_w G\wedge\beta$ is fibre-covector $\wedge\,\beta$. The same factorization therefore gives $n\omega_V^m \wedge (e\sigma_{\mathcal{A}} + \partial_s G\, ds\wedge\beta)$, with $\beta = b(v)\, dv$. Positive coefficients give a nonzero top power. □

Let (q_t) be the comparator isotopy, generated by Q_t with $\iota_{X_{Q_t}}\omega_V = -dQ_t$, and $q = q_1$; in dimension four q_t is the identity on $\{p > P_{\mathrm{supp}}\}$ with $P_{\mathrm{supp}} < h$, so it preserves the fibre disc $\bar{D}(h)$. Let f be a clock, $f = 0$ for $s \le 0$ and $f = 1$ for $s \ge 1$, and let H^* be a function of the moments. For $\lambda \in [0,1]$ define on $\mathcal{A}\times\mathbb{C}^m$

(7)

$$\mathcal{D}_\lambda(s,v,w) = \big(s, v, q_{\lambda f(s)}(w)\big), \qquad Q_s^\lambda = \lambda f'(s)\, Q_{\lambda f(s)}, \qquad G_\lambda(s,w) = f(s)\, H^*\big(p(q_\lambda^{-1}(w))\big),$$

and $\mathcal{D} = \mathcal{D}_1$. Then $\partial_s q_{\lambda f(s)} = X_{Q_s^\lambda}\circ q_{\lambda f(s)}$. The twist changes the fibre identification across the annulus from the identity to q.

Lemma 5.2 (Suspension)**.** *Let σ_0 be a base two-form on $\mathcal{A}$. On $\mathcal{A}\times\mathbb{C}^m$, for $\lambda\in[0,1]$:*

(i) $\mathcal{D}_\lambda^*\omega_V = \omega_V + d\big[(Q_s^\lambda\circ q_{\lambda f(s)})\, ds\big]$;

(ii) with the horizontal one-form

$$\eta_\lambda = (Q_s^\lambda\circ q_{\lambda f(s)})\, ds + f(s)\Big[H^*\big(p(q_\lambda^{-1}q_{\lambda f(s)}(w))\big) - H^*\big(p(w)\big)\Big]\beta,$$

one has $\mathcal{D}_\lambda^*\big(\omega_V+\sigma_0+d(G_\lambda\beta)\big) = \omega_V+\sigma_0+d\big(f(s)H^*(p)\beta\big)+d\eta_\lambda$;

(iii) $\eta_0 = 0$, *η_λ is smooth in (λ,s,v,w), and $\eta_\lambda = 0$ for $s\notin(0,1)$, so it extends by zero to a horizontal one-form supported over $\mathcal{A}$. If the first path ends at a form Ω_1 equal to $\omega_V+\sigma_0+d(f(s)H^*(p)\beta)$ over $\mathcal{A}$ (Lemma S8.5 for $n=2$; for $n\ge3$ see* [Son26a]*), then $\Omega_1 + d\eta_\lambda = \mathcal{D}_\lambda^*\big(\omega_V+\sigma_0+d(G_\lambda\beta)\big)$ over $\mathcal{A}$.*

Proof. (i) Put $\Theta(s,w) = Q_s^\lambda(q_{\lambda f(s)}(w))$ and $q = q_{\lambda f(s)}$; the right side is $\omega_V + d_w\Theta\wedge ds$. The map $d\mathcal{D}_\lambda$ fixes ∂_v, sends a vertical u to $Dq\,u$, and sends ∂_s to $\partial_s + X_{Q_s^\lambda}(q(w))$. Since q is symplectic, both sides agree on vertical pairs. On (∂_s, u) the left side is $\omega_V(X_{Q_s^\lambda}, Dq\,u) = -d_w\Theta(u)$. Pairs involving ∂_v vanish on both sides. (ii) $\mathcal{D}_\lambda$ fixes base points, so it preserves σ_0 and β. Moreover $\mathcal{D}_\lambda^*(G_\lambda\beta) = f(s)H^*(p(q_\lambda^{-1}q_{\lambda f(s)}w))\beta$; add (i). (iii) For $s\le0$, $f = f' = 0$. For $s\ge1$, $f=1$, $f'=0$ and $q_\lambda^{-1}q_{\lambda f(s)} = \mathrm{id}$. The last identity is (ii) with the assumed form of Ω_1. □

Using the fixed map q_λ^{-1} in G_λ makes the bracket vanish identically for $s\ge1$, and it gives $\partial_s G_\lambda = f'(s)H^*(p(q_\lambda^{-1}w))$.

The second path. Put $\Omega_\lambda^{(1)} = \Omega_1 + d\eta_\lambda$; it equals Ω_1 off the annulus. Over the annulus it is $\mathcal{D}_\lambda^*$ of a form whose top power is, by Lemma 5.1(b) with $n=2$, $\sigma_0 = e_h\nu$, $2\big(e_h + b(v)f'(u)H^*(p(q_\lambda^{-1}w))\big)\,\omega_V\wedge du\wedge dv$, positive on $\{p < h+2\delta_m\}$ because $H^*>0$ there. Since $\mathcal{D}_\lambda$ preserves orientation, $\Omega_\lambda^{(1)}$ is an exact symplectic path on this region (Proposition S8.9). Only the forms, not $\mathcal{D}_\lambda$, are glued to the model. The higher-dimensional handle and the packing theorem in a toric model are in the companion paper [Son26a].

Remark 5.3 (Dimension four). For $n=2$ the construction is carried out on the cover, equivariantly under the deck maps, (§S8.1). The clock is $f(u) = S((u-\frac{7}{48})/\frac{1}{24})$, i.e. $S(2s-\frac12)$, and the $k+1$ layer intervals lie inside its ramp. The caps are $h_i(p) = \frac12\big(a_i - p + \sqrt{(a_i-p)^2+\mu_h^2}\big)$ with $\mu_h = g/(16k)$; they lie above the tents and exceed them in total by at most $g/32$, so a comparison gap $\frac{31}{32}g$ leaves a residual of at least $\frac78 g$. The three paths are concatenated into one flow, cut off only in the moment variable, on $\{p < h+2\delta_m\}$, $\delta_m = \min((d_n-h)/4, g/100)$. All three variation primitives vanish as covectors on the marked fibres, so the flow fixes them pointwise and the images avoid $\{Z_0=0\}$ (Lemma S8.14, Propositions S8.16 and S8.17).

6. The four-dimensional model and the return to the ball

This section describes the four-dimensional model, the return to the ball and the margins of the construction; details are in §S6 and §S10. Throughout, the target is $B^\circ(1)$ and $\rho_i = c_i/C$.

6.1. **The cubic model.** The model is the normal disc bundle of Biran's decomposition of $\mathbb{CP}^2$ along a smooth cubic [Bir01], written on the cover $\mathbb{R}^2 \times \mathbb{C}$ of the Fermat torus $\Gamma = \mathbb{C}/(\mathbb{Z} + \zeta\mathbb{Z})$, $\zeta = e^{2\pi i/3}$. Theta functions give a smooth map $Z\colon \mathbb{R}^2 \to \mathbb{C}^3$, periodic up to phase, with $|Z| = 1$ and $Z_0^3 + Z_1^3 + Z_2^3 = 0$. With $\eta_\Gamma = \frac{1}{2\pi} \operatorname{Im} \sum_j \overline{Z_j}\, dZ_j$, $\sigma = d\eta_\Gamma$ and $p = \pi|w|^2$, the model form is

$$\Omega_0 = \omega_V + (1 - 3p)\,\sigma - 3\,dp \wedge \eta_\Gamma, \qquad \int_\Gamma \sigma = 3,$$

so the base area at level p is $3(1 - 3p)$, the fibre interval stops below $p = \frac{1}{3}$, and hence $3c_i < C$. The integrated profile over $[0, h]$ is $3h - \frac{9}{2}h^2 \to \frac{1}{2} = \operatorname{vol} \mathbb{CP}^2$ as $h \uparrow \frac{1}{3}$ (Lemma S6.2).

Theorem 6.1 (Packing in dimension four)**.** *Let $(k, \rho_i, a_i, h, d_n, g, K, q)$ be valid cubic data (Definition S8.1) with $d_n = \frac{1}{2}(h + \frac{1}{3})$: $0 < \rho_i < a_i < h < d_n < \frac{1}{3}$, $0 < g \le 3(1 - 3h)$, and $(K, (q_t))$ a comparison datum for the profile $3(1 - 3p) - \sum_i (a_i - p)_+$ on $[0, h]$ with gap $\frac{31}{32}g$, whose isotopy is the identity near $p = h$. Then there are explicit numbers $\rho_i < \rho_i^+ < \rho_i^{\mathrm{dom}}$ and smooth maps $f_i\colon B^\circ(\rho_i^{\mathrm{dom}}) \to \mathbb{C}^2$ with symplectic derivative, injective on $B(\rho_i^+)$, such that the sets $f_i(B(\rho_i^+))$ are pairwise disjoint and contained in $B(1 - \mu_{\mathrm{ball}})$ for some $\mu_{\mathrm{ball}} \in (0, 1)$.*

This is Theorem S10.23. For $k \ge 10$ equal capacities with $\frac{3}{4} < \rho\sqrt{k} < 1$ the data are given by the linear datum of Proposition 4.4, with $a = t_0/\sqrt{k}$, $h = t_0/3$, $t_0 = (1 + \rho\sqrt{k})/2$, all algebraic in $\sqrt{k}$ and ρ (Remark S10.25). For all other $\rho_i < \frac{1}{3}$ with $\sum_i \rho_i^2 < 1$, including $k = 9$ and unequal capacities, the companion paper [Son26a] constructs valid cubic data from a scalar comparison datum. The embeddings are

$$f_i = b_{\mathrm{ball}} \circ \psi_1^{-1} \circ \mathrm{pr} \circ \tau_1 \circ \iota \circ \mathsf{N} \circ \Phi^{-1} \circ \chi_{\mathcal{A}} \circ \mathcal{D}^{-1} \circ E_i,$$

read from right to left as follows. The layer map E_i, built from an explicit strip map and a unique root, places the ball in the i-th layer over the annulus (Proposition S8.17); $\mathcal{D}$ is the twist and $\chi_{\mathcal{A}}$ the frame chart over the annulus (Section 5); Φ is the handle Moser map, which commutes with the deck maps and fixes the marked fibres (Remark 5.3); $\mathsf{N}(u, v, w) = (Z(u, v), w)$ passes from the cover to unit Fermat representatives; and $\iota, \tau_1, \mathrm{pr}, \psi_1^{-1}, b_{\mathrm{ball}}$ form the return of §6.2. The argument is outlined in §S6.5.

6.2. **The return to the ball.** The handle maps land in $\{p < h\}$ of the model on the cover, and the return carries this region into $\mathbb{CP}^2 \setminus \{Z_0 = 0\}$, which the affine chart b_{ball} identifies with $B^\circ(1)$. It is a deformation to the normal cone, following the scheme of [Ope26], written with explicit flows (Construction S6.4). Let $\mathcal{F}(Z) = Z_0^3 + Z_1^3 + Z_2^3$ and $X_T = \{\mathcal{F}(Z) = TW\}$, with W of weight three: X_0 is the total space of $\mathcal{O}(3)|_E$ over the cubic, and X_1 is a graph over $\mathbb{CP}^2$. The return is $\mathcal{T} = b_{\mathrm{ball}} \circ \psi_1^{-1} \circ \mathrm{pr} \circ \tau_1 \circ \iota \circ \mathsf{N}$: N passes to unit Fermat representatives, ι is an explicit weight-three injection into X_0, τ_1 is the time-one map of an explicit transport field from X_0 to X_1 preserving the restricted forms, pr is the projection of the graph, and ψ_1^{-1} is the inverse of an explicit Moser return removing an exact term from the Fubini–Study form. The denominators of both fields are bounded below on the regions the trajectories visit (Table 1). Then $\mathcal{T}$ pulls ω_0 back to Ω_0 on $\{p < h\}$ off the marked fibres and identifies exactly deck-equivalent points (Theorem S10.21).

The images must avoid the line $L = \{Z_0 = 0\}$, which the ball chart removes. This follows from a symmetry. Let $\mathsf{r}(Z, W) = (\zeta Z_0, Z_1, Z_2, W)$.

(a) r is unitary and preserves $\mathcal{F}$, so it preserves the forms and constraints above; since $\bar{\zeta}^2 = \zeta$ it maps both fields to themselves. Hence ι, τ_1, pr and ψ_1^{-1} commute with r (Lemma S10.17).
(b) Each factor is injective on representatives and commutes with the gauge action $(Z, W) \mapsto (cZ, c^3W)$, so $\mathcal{T}_{\mathrm{u}} = \psi_1^{-1} \circ \mathrm{pr} \circ \tau_1 \circ \iota$ is injective modulo gauge (Lemma S10.18).
(c) For a point $x = (Z, U)$ of the central fibre, $\mathsf{r}x$ is a gauge multiple of x only if $Z_0 = 0$ (Lemma S10.16).

If an output $Y = \mathcal{T}_{\mathrm{u}}(x)$ had $Y_0 = 0$, then $\mathsf{r}Y = Y$, so by (a) and (b) $\mathsf{r}x$ is a gauge multiple of x, and (c) gives $Z_0 = 0$. The handle construction keeps the balls off the marked fibres, the fibres over the points of E with $Z_0 = 0$: the layers lie over the annulus, which misses these points, and the three Moser flows fix the marked fibres pointwise because their variation primitives vanish there (Lemma S8.14). Hence the images avoid L, with a positive margin μ_{ball} to $\partial B^\circ(1)$ on every compact family (Proposition S10.19).

6.3. **Margins.** Each step of the construction leaves a quantitative margin that a later step consumes; Table 1 lists them for the linear branch. The gap $g = 3(1 - t_0) = \frac{3}{2}\eta_\star$ pays for the planar cost ($\le \frac{g}{32}$), the background $e_h = \frac{g}{16}$ and the cap excess ($\le \frac{g}{32}$), leaving a residual of at least $\frac{7}{8}g$.

A worked instance. §S13 takes $k = 10$, $\varepsilon = \frac{1}{10}$ in Theorem A: $R = (k/(1-\varepsilon/4))^{1/4}$, $C = \pi R^2$, $c_i = \pi$, $s_i = (3i - 3, 0, 0, 0)$, so $\rho = \sqrt{39}/20$, $\rho\sqrt{k} = \sqrt{39/40} \in (\frac{3}{4}, 1)$ and the linear datum applies, with $K(p) = (3\sqrt{10} - 9)p$; the supplement lists the resulting constants and margins and the composed Hamiltonian.

7. From embeddings to one Hamiltonian; proof of Theorem A

We join the translations $u \mapsto s_i + u$ to the embeddings f_i of Section 6 by one explicit family $F_{i,t}$ of symplectic embeddings with pairwise disjoint images at every time (§7.2). We then write down a Hamiltonian on $\mathbb{R} \times \mathbb{R}^{2n}$ whose field is the velocity of this family on the moving balls (§7.3). Every stage is a formula, an integral, a matrix function, a maximum over a compact set, a definite description, or the least of finitely many angle indices.

Notation. $\mathbb{B}(y, r)$ and $\mathbb{B}^\circ(y, r)$ are the closed and open Euclidean balls of radius r about $y \in \mathbb{R}^{2n}$, with $\mathbb{B}(r) = \mathbb{B}(0, r)$; thus $B(s, c) = \mathbb{B}(s, \sqrt{c/\pi})$. We write $\omega_0(v, w) = v^{\mathsf{T}} J_0 w$ with $J_0 = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$ in coordinates (x, y), so $X_H = J_0 \nabla H$ and $\iota_{X_H}\omega_0 = dH$. Multiplication by i on $\mathbb{C}^n$, $z_j = x_j + iy_j$, is $J_{\mathbb{C}} = -J_0$. $\mathrm{Sp}(2n) = \{A : A^{\mathsf{T}} J_0 A = J_0\}$, and $\|\cdot\|$ is the Euclidean operator norm.

Two observations drive the construction: the maps $u \mapsto v^{-1} f_i(vu)$, $0 < v \le 1$, are symplectic with disjoint images for a common v; and on the source ball the one-form $\omega_0(\dot{F}, DF\,\cdot)$ is closed, with an explicit radial primitive.

7.1. The embeddings.

Definition 7.1 (Normalisation)**.** Let $C > 0$ and $c \in (0, \infty)^k$. The *normalised capacities* are $\rho_i = c_i/C$. For a map $f^\natural$ defined on a subset of $\mathbb{R}^{2n}$, its *physical rescaling* is $f(u) = \sqrt{C}\, f^\natural(u/\sqrt{C})$.

The dilation $\delta(z) = z/\sqrt{C}$ maps $B(s, c)$ onto $B(s/\sqrt{C}, c/C)$ and $B^\circ(C)$ onto $B^\circ(1)$. Definition 2.2 is homogeneous, so (n, k, C, c) is admissible iff $(n, k, 1, \rho)$ is. Moreover $f = \delta^{-1} \circ f^\natural \circ \delta$ has $Df(u) = Df^\natural(u/\sqrt{C})$, so it is symplectic, injective or of disjoint images exactly when $f^\natural$ is.

In the *linear branch*, $n = 2$, $k \ge 10$, equal normalised capacities $\rho_i = \rho$ and $\frac{3}{4} < \rho\sqrt{k} < 1$, Construction S11.1 defines *normalised embeddings* $f_i^\natural$, an enlarged closed capacity ρ_i^+ and an open domain capacity ρ_i^{dom}, with $\rho_i < \rho_i^+ < \rho_i^{\mathrm{dom}}$, from the cubic model with the linear datum. These

Margin	Definition; dependence	Used for	Source
volume defect	$\eta_\star = 1 - \rho\sqrt{k}$; input	$\eta_\star = (1-t_0) + \sqrt{k}(a-\rho)$, $\frac{\eta_\star}{2}$ each	hypothesis
capacity slack $a-\rho$	$\frac{\eta_\star}{2\sqrt{k}}$	half to $\bar\rho$; strip excess $\frac{\delta}{2}$; $\frac{\delta}{4}$ left below h_i	Rem. S10.25
ball enlargement $\rho < \rho^+ < \rho^{\mathrm{dom}}$	$(\rho\varrho_3^2, \rho\varrho_4^2)$; ρ, a	closed source $B(\rho^+)$, open domain $B^\circ(\rho^{\mathrm{dom}})$	Rem. S10.24, Prop. S8.17
fibre height $1-3h$	$1-t_0 = \frac{\eta_\star}{2}$	$g \le H_{\mathrm{b}}(h) = 3(1-3h)$; transfer margins	Rem. S10.25
gap g	$3(1-t_0)$	all handle margins below	Prop. 4.4
planar tolerance e_q	$\min(\frac{h}{100}, \frac{g}{128c_s})$	$\Lambda(p(q(z))) \le \pi\lvert z\rvert^2 + 4e_q$; cost $4c_se_q \le \frac{g}{32}$	Def. S3.4, Thm. S5.34, Lem. S3.5
comparison gap $\ge \frac{31}{32}g$	g minus the planar cost	background, cap excess, residual	Prop. 4.4
background $e_h = \frac{g}{16}$	$H^* = H_{\mathrm{b}} - e_h \ge \frac{15}{16}g$ on $[0,h]$	positive form $e_h\,du\wedge dv$ on the annulus	Rem. 5.3, Lem. S8.3(b)
cap width $\mu_h = \frac{g}{16k}$	$h_i = \frac{1}{2}(a_i - p + \sqrt{(a_i-p)^2+\mu_h^2})$	cap excess $\sum_i (h_i - (a_i-p)_+) \le \frac{g}{32}$	Rem. 5.3, Lem. S8.3(a)
residual $\frac{7}{8}g$	$\frac{31}{32}g - e_h - \frac{g}{32}$	positive remainder layer R^*	Lem. S8.3(c)
planar support $h - P_{\mathrm{supp}}$	isotopy data of Def. S4.11; $P_{\mathrm{supp}} > h - \frac{31}{8}e_q$	all flows preserve p above P_{supp}	Lem. S4.12, positivity
safe region $\{p < h + 2\delta_m\}$	$\delta_m = \min(\frac{d_n - h}{4}, \frac{g}{100})$; cutoff on $[h+\frac{3}{2}\delta_m, h+\frac{7}{4}\delta_m]$	$H^* > \frac{15}{16}g - 18\delta_m$; Moser identity on $\{p \le h+\delta_m\}$	Lem. S8.3(d)
base speed C_b	maximum over a compact fundamental set	completeness of the handle flow	positivity only
transfer margins	$d_n - h = \frac{1}{2}(\frac{1}{3}-h)$, $1-3d_n = \frac{1}{2}(1-3h)$; h	ι needs $p < d_n$; $A > 1-3d_n$; $w_* = \frac{h}{d_n - h}$	explicit
transport denominator	$N_{\mathrm{den}} \ge c_* = \min(\frac{3}{2}, \frac{T_0^2}{d_n})$; d_n, w_*	cutoff at $\frac{c_*}{2}$, inactive on $\mathcal{C}_*$	Lem. S10.9
return denominator	$A_s + B_sG \ge 1 - \frac{3}{2}d_n > \frac{1}{2}$; d_n	cutoff at half the bound, inactive	Lem. S10.12
target boundary μ_{ball}	$\frac{1}{2}\min_{\mathcal{K}}(1-\pi\lvert z\rvert^2) = \frac{1}{2}\min\lvert Y_0\rvert^2$	compactness of $\bigcup_i f_i(B(c_i')) \subset B^\circ(C)$	Prop. S10.19, positivity only
radii $r_i < r_{1,i} < \cdots < r_{4,i} < r_i'$	$r_{\ell,i} = r_i + \frac{\ell}{5}\min(\frac{1}{4}\text{centre gap}, r_i' - r_i)$	cutoff collar, disjoint closed balls, open domain	Lem. 7.5
isotopy scale τ	$\min(\frac{1}{2}, \frac{\delta_c}{2Lr_{\mathrm{tot}}})$; L a maximum over compact sets	separation $\lvert Y_i - Y_j\rvert \ge 4Lr_{\mathrm{tot}}$	Lem. S12.5, positivity only

TABLE 1. Margins of the construction in the linear branch, in the order in which they are used. The table is an index into the supplement: symbols not defined in the main text are defined at the cited statements. "Explicit" means that the margin is a closed formula in the inputs; "positivity only" means that the proof shows the number is positive without bounding it.

capacities are closed formulas in ρ built from arithmetic and square roots. The constructions of this section other than the embeddings hold in every dimension.

Definition 7.2 (Embedding datum)**.** Let $n \geq 1$, $k \geq 1$, $C > 0$ and $c \in (0,\infty)^k$. An *embedding datum* for (n,k,C,c) consists of numbers $c_i', \bar{c}_i$ with $c_i < c_i' < \bar{c}_i$ and maps $f_i \colon B^\circ(\bar{c}_i) \to \mathbb{R}^{2n}$ such that

(i) f_i is C^∞ and $Df_i(u) \in \mathrm{Sp}(2n)$ for all $u \in B^\circ(\bar{c}_i)$;
(ii) f_i is injective on $B(c_i')$;
(iii) the sets $f_i(B(c_i'))$, $1 \leq i \leq k$, are pairwise disjoint and contained in $B^\circ(C)$.

Proposition 7.3 (The embeddings)**.** *Let $k \geq 10$, $C > 0$ and $c_1 = \cdots = c_k = c > 0$ with $\rho = c/C$ in the linear branch, $\frac{3}{4} < \rho\sqrt{k} < 1$, and let $f_i^\natural$, ρ_i^+ and ρ_i^{dom} be given by Construction S11.1. Put*

$$c_i' = C\rho_i^+, \qquad \bar{c}_i = C\rho_i^{\mathrm{dom}}, \qquad f_i(u) = \sqrt{C}\, f_i^\natural(u/\sqrt{C}).$$

Then $(c_i', \bar{c}_i, f_i)$ is an embedding datum for $(2,k,C,c)$, and $\bigcup_i f_i(B(c_i'))$ is a compact subset of $B^\circ(C)$.

Proof. Theorem S10.23, for the valid cubic data of the equal case, gives smoothness and the symplectic identity on $B^\circ(\rho_i^{\mathrm{dom}}) \supset B(\rho_i^+)$. They also give injectivity, pairwise disjointness and target containment on $B(\rho_i^+)$. Physical rescaling preserves these properties. The union is a finite union of continuous images of compact sets, hence compact. □

7.2. **A common symplectic isotopy.** Throughout this subsection, fix an embedding datum $(c_i', \bar{c}_i, f_i)$ for (n,k,C,c) and centres s_i with $B(s_i,c_i)$ pairwise disjoint. All proofs are in §S12.

Definition 7.4 (Radii)**.** Put $r_i = \sqrt{c_i/\pi}$, $r_i' = \sqrt{c_i'/\pi}$, $\bar{r}_i = \sqrt{\bar{c}_i/\pi}$,

$$\delta_{\mathrm{sep}} = \begin{cases} \min_{i\neq j}\big(|s_i - s_j| - r_i - r_j\big), & k \geq 2, \\ 1, & k = 1, \end{cases}$$

$$\delta_i = \min\big(\tfrac{\delta_{\mathrm{sep}}}{4},\ r_i' - r_i\big), \qquad r_{\ell,i} = r_i + \tfrac{\ell}{5}\delta_i \quad (\ell = 1,\dots,4).$$

Lemma 7.5 (Radii)**.** *$\delta_{\mathrm{sep}} > 0$, $\delta_i > 0$, $r_i < r_{1,i} < r_{2,i} < r_{3,i} < r_{4,i} < r_i' < \bar{r}_i$, and*

$$|s_i - s_j| > r_{3,i} + r_{3,j} \qquad (i \neq j). \tag{8}$$

Consequently f_i is C^∞ with symplectic derivative on $\mathbb{B}^\circ(\bar{r}_i)$, injective on $\mathbb{B}(r_{4,i})$, and the sets $f_i(\mathbb{B}(r_{4,i}))$ are pairwise disjoint and contained in $\mathbb{B}^\circ(R_0)$, $R_0 = \sqrt{C/\pi}$: the data form an isotopy input (Definition S12.1).

Proof. If $|s_i - s_j| \leq r_i + r_j$, then $s_i + \frac{r_i}{r_i+r_j}(s_j - s_i) \in \mathbb{B}(s_i,r_i) \cap \mathbb{B}(s_j,r_j)$, a contradiction; hence $\delta_{\mathrm{sep}} > 0$. Since $c_i' > c_i$, we have $\delta_i > 0$, and so $r_{4,i} < r_i + \delta_i \leq r_i' < \bar{r}_i$. For $i \neq j$, $r_{3,i} + r_{3,j} \leq r_i + r_j + \frac{3}{10}\delta_{\mathrm{sep}} < |s_i - s_j|$. The rest follows from Definition 7.2, since $\mathbb{B}^\circ(\bar{r}_i) = B^\circ(\bar{c}_i)$, $\mathbb{B}(r_{4,i}) \subset B(c_i')$ and $B^\circ(C) = \mathbb{B}^\circ(R_0)$. □

The radii play three roles. $\mathbb{B}^\circ(r_{4,i})$ is the source domain of every stage. The images of the balls $\mathbb{B}(r_{3,i})$ are kept pairwise disjoint. The cutoff of §7.3 interpolates between $r_{1,i}$ and $r_{2,i}$. Table 1 lists all margins of the construction.

The isotopy has four kinds of stages:

- *Route* (Definition S12.10). The round balls are translated to $Y_i = y_i/\tau$, where $y_i = f_i(0)$. Each ball rises in x_1 to a level ranked by its starting height, then in y_1 to a level ranked by its target height. All balls then move straight to their targets, and the offsets are removed in reverse order. Since ranks are compatible with heights, the balls $\mathbb{B}(r_{3,i})$ stay disjoint (Lemma S12.11).

- *Linear stage.* $Y_i + M_{i,\sigma}u$, where $\sigma \mapsto M_{i,\sigma}$ is an explicit smooth path in $\mathrm{Sp}(2n)$ from I to $A_i = Df_i(0)$. It is built from a polar decomposition, a Cayley transform and a shear (Definition S12.2).
- *Shape stage.* $Y_i + \kappa_{i,\tau\sigma}(u)$, where $\kappa_{i,v}(u) = \int_0^1 Df_i(\vartheta v u)\, u\, d\vartheta$. It runs from $Y_i + A_i u$ to $\tau^{-1} f_i(\tau u)$.
- *Tail.* $T_{i,v}(u) = v^{-1} f_i(vu)$ for v from τ to 1, with $T_{i,1} = f_i$. Disjointness is preserved because one dilation acts on all images (Lemma S12.9).

The scale $\tau \in (0, \frac{1}{2}]$ is given by a formula in two quantities. One is a bound L for $\|Df_i\|$ and $|M_{i,\sigma}u|$ on $\mathbb{B}(r_{3,i})$, a maximum of a continuous function over a compact set. The other is the minimal distance of the points y_i. The formula makes the balls $\mathbb{B}(Y_i, Lr_{\mathrm{tot}})$ pairwise disjoint (Definition S12.4). The stages occupy the slots $[\frac{m}{8}, \frac{m+1}{8}]$ and are joined by flat clocks $S(8t - m)$, where S is the flat step. On each slot the defining sum $F_{i,t}$ reduces to one stage (Construction S12.12 and Lemma S12.13).

Proposition 7.6 (The common symplectic isotopy)**.** *(a) $(t, u) \mapsto F_{i,t}(u)$ is C^∞ on $\mathbb{R} \times \mathbb{B}^\circ(r_{4,i})$; $F_{i,0}(u) = s_i + u$, $F_{i,1} = f_i$; $F_{i,t} = F_{i,0}$ for $t \le 0$ and $F_{i,t} = F_{i,1}$ for $t \ge 1$;*

(b) $D_u F_{i,t}(u) \in \mathrm{Sp}(2n)$ for all t and $u \in \mathbb{B}^\circ(r_{4,i})$;

(c) each $F_{i,t}$ is injective on $\mathbb{B}(r_{3,i})$, and for every t the sets $F_{i,t}(\mathbb{B}(r_{3,i}))$, $1 \le i \le k$, are pairwise disjoint;

(d) $F_{i,t}(\mathbb{B}(r_{3,i})) \subset \mathbb{B}(R_F)$ for all i, t, for the explicit radius R_F of Proposition S12.14.

Proof. By Lemma 7.5 the data form an isotopy input, and Proposition S12.14 gives (a)–(d). The key point is that (b)–(d) need only be checked for one stage per slot. Route slots are translations, with disjoint images by the rank argument. Linear and shape slots have symplectic derivatives $M_{i,\sigma}$ and $Df_i(\tau\sigma u)$, and their images lie in the disjoint balls $\mathbb{B}(Y_i, Lr_{\mathrm{tot}})$. The tail slot has derivative $Df_i(vu)$, and its images are disjoint because the single dilation $y \mapsto v^{-1}y$ is applied to the disjoint sets $f_i(v\mathbb{B}(r_{3,i})) \subset f_i(\mathbb{B}(r_{4,i}))$. □

7.3. **The Hamiltonian.** Let F be the isotopy of Construction S12.12, with radii $r_{1,i} < r_{2,i} < r_{3,i} < r_{4,i}$; only properties (a)–(d) of Proposition 7.6 are used.

Definition 7.7 (Source one-form and primitive)**.** For $t \in \mathbb{R}$, $u \in \mathbb{B}^\circ(r_{4,i})$ and $w \in \mathbb{R}^{2n}$,

$$\mathfrak{a}_{i,t}(u)[w] = \omega_0\big(\dot F_{i,t}(u), D_u F_{i,t}(u) w\big), \qquad \mathfrak{h}_{i,t}(u) = \int_0^1 \mathfrak{a}_{i,t}(\vartheta u)[u]\, d\vartheta,$$

where $\dot F = \partial_t F$.

Lemma 7.8 ($\mathfrak{a}_{i,t}$ is closed)**.** *The bilinear form $(w, w') \mapsto D_u \mathfrak{a}_{i,t}(u)[w][w']$ is symmetric; that is, $d_u \mathfrak{a}_{i,t} = 0$.*

Proof. With $F = F_{i,t}$ and $\partial_l = \partial/\partial u_l$, the second-derivative terms cancel and $\partial_l \dot F = \partial_t \partial_l F$, so the $du_l \wedge du_m$ coefficient of $d_u \mathfrak{a}_{i,t}$ is $\partial_t[\omega_0(\partial_l F, \partial_m F)]$. This vanishes because $\omega_0(\partial_l F, \partial_m F) = \omega_0(e_l, e_m)$ by Proposition 7.6(b). □

Lemma 7.9 (Radial primitive)**.** *$(t, u) \mapsto \mathfrak{h}_{i,t}(u)$ is C^∞ on $\mathbb{R} \times \mathbb{B}^\circ(r_{4,i})$, $\mathfrak{h}_{i,t}(0) = 0$, $d_u \mathfrak{h}_{i,t} = \mathfrak{a}_{i,t}$, and $\mathfrak{h}_{i,t} \equiv 0$ for $t \notin (0, 1)$.*

Proof. Differentiating under the integral and using Lemma 7.8 gives $d_u \mathfrak{h}_{i,t}(u)[w] = \int_0^1 \frac{d}{d\vartheta}(\vartheta\, \mathfrak{a}_{i,t}(\vartheta u)[w])\, d\vartheta = \mathfrak{a}_{i,t}(u)[w]$. For $t \notin (0, 1)$, $F_{i,\cdot}(u)$ is constant near t, so $\dot F_{i,t} = 0$ and $\mathfrak{a}_{i,t} = 0$. □

Definition 7.10 (Source cutoff and the Hamiltonian of an embedding datum)**.** Let $\chi_i = \chi_{r_{1,i}, r_{2,i}}$ be the radial cutoff $\chi_{r,r'}(u) = 1 - S\big((|u|^2 - r^2)/(r'^2 - r^2)\big)$, so $\chi_i = 1$ on $\mathbb{B}(r_{1,i})$ and $\chi_i = 0$

outside $\mathbb{B}^\circ(r_{2,i})$ (Lemma S2.7). Put $H_i(t,y) = \chi_i(u)\,\mathfrak{h}_{i,t}(u)$ if u is the unique point of $\mathbb{B}(r_{3,i})$ with $F_{i,t}(u) = y$, and $H_i(t,y) = 0$ if there is no such unique point; and $H_{\mathfrak{E}} = \sum_{i=1}^k H_i$, where $\mathfrak{E}$ denotes the embedding datum together with the centres. The unique preimage is a definite description (Definition 2.4).

By Proposition 7.6(c), the first case applies exactly when $y \in F_{i,t}(\mathbb{B}(r_{3,i}))$.

Lemma 7.11 (Smoothness and support)**.** *Each H_i is C^∞ on $\mathbb{R}\times\mathbb{R}^{2n}$; $H_i(t,\cdot)$ vanishes outside $F_{i,t}(\mathbb{B}(r_{2,i}))$ and vanishes identically for $t \notin (0,1)$. Hence $H_{\mathfrak{E}}$ is C^∞, vanishes for $t \notin (0,1)$, and $H_{\mathfrak{E}}(t,\cdot)$ vanishes outside the compact set $K_H = \bigcup_i \bigcup_{t\in[0,1]} F_{i,t}(\mathbb{B}(r_{2,i})) \subset \mathbb{B}(R_F)$.*

Proof. The map $E_i(t,u) = (t, F_{i,t}(u))$ is a smooth injective local diffeomorphism on $\mathbb{R}\times\mathbb{B}(r_{3,i})$. Hence $H_i = (\chi_i\mathfrak{h}_i)\circ E_i^{-1}$ is smooth on $E_i(\mathbb{R}\times\mathbb{B}^\circ(r_{3,i}))$ and vanishes off the closed set $E_i(\mathbb{R}\times\mathbb{B}(r_{2,i}))$ (closed by Proposition 7.6(a)), which gives smoothness (Lemma S12.16). The time support follows from Lemma 7.9. Compactness of K_H holds because F is constant in t outside $[0,1]$, and $K_H \subset \mathbb{B}(R_F)$ by Proposition 7.6(d). □

Lemma 7.12 (The field on the moving balls)**.** *For all $t\in\mathbb{R}$, $u\in\mathbb{B}^\circ(r_{1,i})$ and $w\in\mathbb{R}^{2n}$,*

$$d_y H_{\mathfrak{E}}(t, F_{i,t}(u))[w] = \omega_0\big(\dot F_{i,t}(u), w\big), \qquad \text{hence} \qquad X_{H_{\mathfrak{E}}}(t, F_{i,t}(u)) = \dot F_{i,t}(u).$$

Proof. For x near u, $H_i(t, F_{i,t}(x)) = \mathfrak{h}_{i,t}(x)$. Differentiating with Lemma 7.9 gives $d_yH_i \circ D_uF_{i,t}(u) = \omega_0(\dot F_{i,t}(u), D_uF_{i,t}(u)\,\cdot)$, where $D_uF_{i,t}(u)$ is invertible. For $j\neq i$, $H_j(t,\cdot)$ vanishes on the open set $F_{i,t}(\mathbb{B}^\circ(r_{3,i}))$ by Proposition 7.6(c). Nondegeneracy of ω_0 then gives the claim. □

Lemma 7.13 (Integral curves)**.** *Let $H = H_{\mathfrak{E}}$. Through every point of $\mathbb{R}^{2n}$ passes an integral curve $\gamma\colon\mathbb{R}\to\mathbb{R}^{2n}$ of X_H with $\gamma(0)$ that point. If $|u| < r_{1,i}$ and $\gamma\colon[0,1]\to\mathbb{R}^{2n}$ is an integral curve of X_H with $\gamma(0) = s_i + u$, then $\gamma(t) = F_{i,t}(u)$ for all $t\in[0,1]$.*

Proof. $X_H = J_0\nabla H$ is smooth and supported in $[0,1]\times K_H$ (Lemma 7.11), so it is globally Lipschitz and Picard–Lindelöf applies. By Lemma 7.12, $t\mapsto F_{i,t}(u)$ is the integral curve from $s_i + u$. □

Theorem 7.14 (The Hamiltonian realises the isotopy)**.** *Let $H = H_{\mathfrak{E}}$. The flow ϕ_H^t is defined for all $t\in\mathbb{R}$ and consists of symplectic diffeomorphisms of $\mathbb{R}^{2n}$, and*

$$\phi_H^t(s_i + u) = F_{i,t}(u) \qquad (u\in\mathbb{B}(r_i),\ t\in\mathbb{R},\ 1\le i\le k).$$

In particular $\phi_H^1(B(s_i,c_i)) = f_i(B(c_i))$, and these sets are pairwise disjoint and contained in $B^\circ(C)$.

Proof. The flow is global by the proof of Lemma 7.13, and it is symplectic because it is Hamiltonian. The formula holds by Lemma 7.12, since $\mathbb{B}(r_i)\subset\mathbb{B}^\circ(r_{1,i})$. At $t=1$, $F_{i,1} = f_i$ on $\mathbb{B}(r_i) = B(c_i)\subset B(c_i')$. □

Proposition 7.15 (Hamiltonian of an embedding datum)**.** *Let $n\ge 1$, let $(c_i', \bar c_i, f_i)$ be an embedding datum for (n,k,C,c), and let $s_1,\dots,s_k$ be centres such that the closed balls $B(s_i,c_i)$ are pairwise disjoint. Then the Hamiltonian $H_{\mathfrak{E}}$ of Definition 7.10 satisfies* $\mathrm{Pack}(n,k,C,c,s)$.

Proof. Condition (a) of Definition 2.1 is Lemma 7.11 with $K = \mathbb{B}(R_F)$. For (b), let $z = s_i + u$ with $|u|\le r_i < r_{1,i}$. Lemma 7.13 gives a curve on $\mathbb{R}$ through z, and every integral curve γ on $[0,1]$ from z satisfies $\gamma(1) = f_i(u)\in f_i(B(c_i'))\subset B^\circ(C)$. □

Definition 7.16 (The Hamiltonian $H[2,k,C,c,s]$ of the linear branch)**.** Let $k\ge 10$, $C>0$, $c_1 = \cdots = c_k = c$ with $\frac34 < (c/C)\sqrt{k} < 1$, and let $s_1,\dots,s_k\in\mathbb{R}^4$ be centres with the closed balls $B(s_i,c)$ pairwise disjoint. Let $\mathfrak{E}$ be the embedding datum of Proposition 7.3, together with the centres s_i, and put $H[2,k,C,c,s] = H_{\mathfrak{E}}$ (Definition 7.10).

Remark 7.17 (The general definition). On these data the case distinction of [Son26a] selects this embedding datum, so $H[2,k,C,c,s]$ agrees with the general Hamiltonian there and with `General.packingHamiltonian` of the Lean development; $H_{k,\varepsilon}$ is `General.epochHamiltonianQ`.

7.4. **Proof of Theorem A and Corollary C.**

Proof of Theorem A and Corollary C. Let $k \geq 10$, $s_i = (3i-3,0,0,0)$, $c_i = \pi$, and let R be as in Theorem A, or as in Corollary C with $\varepsilon < \frac{7}{8}$; put $C = \pi R^2$ and $\rho = \pi/C = R^{-2}$. The closed unit balls $B(s_i,\pi)$ are disjoint, since their centres are at distance at least 3. Since $R^4 > k \geq 10$, $k\pi^2 < C^2$ and $3\pi < C$, so $(2,k,C,(\pi,\dots,\pi))$ is admissible, and $\rho\sqrt{k} = \sqrt{1-\varepsilon/4}$, respectively $\sqrt{1-\varepsilon/2}$, lies in $(\frac{3}{4},1)$: the data lie in the linear branch. Let $H = H[2,k,C,(\pi,\dots,\pi),s]$. By Proposition 7.3 and Lemma 7.11, H is smooth and vanishes for $t \notin (0,1)$ and outside $\mathbb{R}\times\mathbb{B}(R_F)$. By Proposition 7.15, it satisfies $\mathrm{Pack}(2,k,C,(\pi,\dots,\pi),s)$. By Theorem 7.14, the images $\phi_H^1(B(s_i,\pi)) = f_i(B(\pi))$ are pairwise disjoint and lie in the compact set $\bigcup_i f_i(B(c_i')) \subset B^\circ(C)$, the open Euclidean ball of radius R. Finally $k/R^4 = 1-\varepsilon/4$, respectively $1-\varepsilon/2$. □

Each entry of Table 2 is an operation of Definition 2.5. The only search is the least of $3 = n+1$ angle indices, and one definition serves all (k,ε); so $H_{k,\varepsilon}$ is defined at level L1.

Ingredient	Operation of Definition 2.5	Where shown
capacities ρ_i^+, ρ_i^{dom}	arithmetic, square roots	Cons. S11.1
model embeddings $f_i^{\natural}$	model formulas, compact flow parameters	Thm. 6.1
c_i', $\bar{c}_i$, f_i	arithmetic; dilation	Prop. 7.3
radii $r_{\ell,i}$	square roots; finite minima; arithmetic	Lem. 7.5
linear path $M_{i,\sigma}$	positive square root (definite description); adjugate inverses; least of $3 = n+1$ angle indices; sin, cos	Lem. S12.3
L, δ_c, r_{tot}, τ, Y_i	maxima over compact sets; finite minima; arithmetic	Lem. S12.5
route z_i	ranks (counts of propositions); arithmetic	Lem. S12.11
shape, tail, clocks	integral over $[0,1]$; composition; flat step	Lem. S12.8, S12.9
$H_{\mathfrak{e}}$	integral over $[0,1]$; radial cutoff; definite description of the preimage	Def. 7.10, Lem. 7.11

TABLE 2. The construction of $H[2,k,C,c,s]$ in the linear branch by the operations of level L1.

8. Formal verification

The Lean 4 development [Son26c, dMU21] uses toolchain `v4.34.1`, Mathlib [The20] at the same tag, and the unmodified library accompanying [Ope26], which contributes only lemmas and generic differential-geometric operators (exterior derivative, chart differential); the Hamiltonian and algorithms are defined in the development, and none of that library's packing constructions or existence results is used. In `verification/`, `Problem.lean` defines the problem from Mathlib alone, `Algorithm.lean` exposes the construction and `Statement.lean` states its correctness; the files for the algorithms are described in [Son26a]. Theorems there have no namespace; other names are relative to `SymplecticPacking`.

Statement	Lean
Theorem A	`four_dim_theorem`, `four_dim_data`, with `General.epochRadiusQ`, `General.epochHamiltonianQ`; `main_time_support`, `main_support_radius`, `flow_images_disjoint`
Proposition B	`four_dim_stability`
Corollary C	`epoch_packing` ($k \geq 9$); closed-ball form `epoch_theorem_ten` ($k \geq 10$), `epoch_theorem` ($k \geq 9$)
Definitions 2.1, 2.2	`Problem.PackingSpec`, `Problem.Admissible`
level L1	`#check_explicit General.packingHamiltonian` (a metaprogram)
embeddings	`Branch2.embeddingsEq_spec` (linear datum)

TABLE 3. Formal counterparts of the main statements (full statements, up to the differences listed below; the Lean statements for Corollary C are stronger). The statements of the companion paper [Son26a] are proved in the same development.

The statement. Phase space is `Problem.E` n = `EuclideanSpace` $\mathbb{R}$ (`Fin` $(2n)$), coordinates $j < n$ and $n+j$ being x_{j+1} and y_{j+1}. `omega0` is $\omega_0 = \sum_j dx_j \wedge dy_j$; `hamVF` is $X_H = (\partial_y H, -\partial_x H)$, so $\iota_{X_H}\omega_0 = dH$; `capBall` $s\ c = \{z : \pi\|z - s\|^2 \leq c\}$ (strict version `capBallOpen`). `Admissible`, `PackingSpec`, `BallsDisjoint` are field by field Definitions 2.2, 2.1 and the disjointness hypothesis; `PackingSpec` quantifies over every integral curve on $[0, 1]$, so is independent of a chosen solution. `EpochSpecR` is the problem as posed: closed unit balls about `epochCenter` $i = (3i, 0, 0, 0)$, $0 \leq i < k$ (the centre s_{i+1}), and a closed target of radius R.

Every theorem concerns the defined term: `Algorithm.lean` shows by `rfl` that `General.packingHamiltonian` is `HamEnd.hamiltonianOfEmbeddings` applied to `General.selectEmbeddings` (the case distinction of Construction S11.1), and that the Epoch Hamiltonian is its instance with $C = \pi R^2$. Differences from the paper: `four_dim_stability` also records $R < R'$; `four_dim_theorem` contains both `EpochSpecR` and the packing specification. The Lean statements for Corollary C are strictly stronger: `epoch_packing`, and its closed-ball form `epoch_theorem` (Epoch's statement with a closed target of radius R), hold for all $k \geq 9$ and $\varepsilon \in (0, 1)$; `epoch_theorem_ten` is the closed-ball form for $k \geq 10$ and all $\varepsilon \in (0, 1)$. For $k = 9$ or $\varepsilon \geq \frac{7}{8}$ their Lean proofs use the scalar branch, which is proved in the companion paper [Son26a]. Theorem A and Corollary C themselves use only the linear branch.

What is verified. The kernel checks the theorems of `Statement.lean` and all dependencies. The trusted base is the kernel, the definitions in `Problem.lean`, and the axioms `propext`, `Classical.choice`, `Quot.sound`; the build fails if any theorem's transitive axiom set contains anything else, in particular `sorryAx`.

Other statements. Most statements of the text carry source tags naming their Lean counterparts; the tags are indexed in `verification/PAPER-INDEX.md`, and the build checks that every tagged name exists and depends only on the three axioms. A tag means either *full statement verified* (up to the differences of form above), or *needed consequence verified*, i.e. only the version the construction uses, with restrictions recorded in Appendix SA.5 (among them return identities along smooth unit-vector families, transfer properties along the constructed embeddings, and the coarser Hamiltonian support bound); an unqualified tag is a reference only. In the main text only Lemma 7.11 is verified in the needed form. These labels are descriptive; only the Lean statements are kernel-checked, and the main correctness theorems concern the defined Hamiltonian and depend on every step of the construction.

Explicitness and uniformity. Since Mathlib's reals use classical choice, axioms cannot distinguish a formula from an arbitrary existential witness. Explicitness (L1) is therefore checked by an unverified metaprogram; definite descriptions (Definition 2.4) are allowed only after unique existence.

`#check_explicit` traverses the non-proof dependency closure of a definition (project, vendored and Mathlib) and accepts only case distinctions on propositions, least-element searches over integers, rationals and finite data with a non-classical decision procedure, choice inside an allow-list of Mathlib definitions (`SymplecticPacking/Explicit/Whitelist.lean`), and definite descriptions after unique existence over types without a discrete factor. The 26 definite descriptions reached from `General.packingHamiltonian` (inverses, roots, critical points, linear solves, matrix square roots, ODE solutions, extrema, measures) are listed in Appendix SA. It is applied to `General.packingHamiltonian`, the Epoch Hamiltonians, the support radius, the case embeddings and the algorithm terms. The metaprogram for uniformity, `#check_constructive`, is described in [Son26a]. `test/ExplicitTest.lean` contains terms that must be rejected, among them a search for the least index whose enumerated Hamiltonian satisfies the specification and the same search wrapped as a unique pair in $\mathbb{N} \times \mathbb{R}$.

Independence. `verification/Audit/Independence.lean` runs `#assert_construction_independence`, which traverses types, definitions and proof bodies of the packing, stability, flow, corollary and algorithm theorems and fails if they reach the vendored packing predicate or existence theorems, the holomorphic-line and two-ball obstruction proofs, or the vendored curve-analysis modules. The vendored declarations used are listed in Appendix SA.4.

Reproduction. With Elan, Node.js and Tectonic installed, run `lake exe cache get` and `bash scripts/verify.sh`, which builds the development with all axiom, explicitness, uniformity and independence gates, runs the regression tests and paper-tag check, and compiles the papers of the release; every stage is fatal. Afterwards `bash scripts/check_kernel.sh` replays the proof closure of `verification/Kernel.lean` into a fresh environment with the pinned `leanchecker`. The development is released under Apache-2.0, with a reviewer guide `REVIEW.md`.

Authorship and the role of AI

Liang Song is the sole author and is responsible for the manuscript and its release. The mathematics, the Lean formalisation and the text were produced by AI systems, including GPT-6.1 Sol (OpenAI), Claude Opus 5.5 (Anthropic) and other models, working in a research environment designed and directed by Liang Song, who chose the problem, assembled the literature and context given to the systems, designed the verification pipeline and made the editorial decisions. The scope of the formal verification is described in Section 8.

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